

Erdős Problem #1055

EP-1055

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

A prime p is in class 1 if the only prime divisors of $p + 1$ are 2 or 3. In general, a prime p is in class r if every prime factor of $p + 1$ is in some class $\leq r - 1$, with equality for at least one prime factor.

Are there infinitely many primes in each class? If p_r is the least prime in class r , then how does $p_r^{1/r}$ behave?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>